

Parameter Transfers for Warm-Started QAOA

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Abstract: Variational Quantum Algorithms (VQAs) run shallow parameterized quantum circuits on quantum devices and are thus suitable for current limited hardware. However, adverse effects like barren plateaus and local optima may hinder the classical parameter optimization. Warm-starting techniques attempt to alleviate such problems by utilizing pre-computed approximations or known solutions to initialize VQAs. In this work, we analyze a combination of two different kinds of warm-starts, based on biased initial states and parameter transfers, respectively. In particular, we investigate a warm-started variant of the Quantum Approximate Optimization Algorithm (WS-QAOA) applied to the MAXCUT problem and analyze the transferability of optimized parameter values between random graphs. We leverage their decomposition into subgraphs and regularities among these subgraphs to facilitate parameter transfers for WS-QAOA and demonstrate a transfer for a random graph.

1 INTRODUCTION

Quantum computing is an emerging field in computer science with promising applications in various domains, such as chemistry and optimization (Cao et al., 2018; Harrigan et al., 2021). However, this potential cannot be fully realized yet, as current *Noisy Intermediate-Scale Quantum (NISQ)* devices are error-prone and the size of quantum circuits that can be executed successfully is limited (Preskill, 2018; Leymann and Barzen, 2020). Errors can accumulate during lengthy computations on quantum devices, making circuit depth a particular issue. Thus, circuits executed on current hardware should be shallow. Until fault-tolerant quantum computers become available, *Variational Quantum Algorithms (VQAs)* bridge the gap and make use of current, limited quantum hardware (Cerezo et al., 2021).

VQAs resort to shallow parameterized quantum circuits, frequently referred to as *Ansätze*, that are executed on the quantum computer, while their parameters are adjusted by an optimizer that is executed on classical hardware. A prominent example of such algorithms is the *Quantum Approximate Optimization Algorithm (QAOA)* (Farhi et al., 2014), which is designed to solve various combi-

natorial optimization problems like the MAXCUT, MINIMUMVERTEXCOVER, and TRAVELINGSALESPERSON problem (Farhi et al., 2014; Zhang et al., 2022; Ruan et al., 2020).

However, VQAs may run into problems like barren plateaus and local minima (Wang et al., 2021; Bittel and Kliesch, 2021), which are detrimental effects in the optimization that may lead to slow convergence of the algorithm. One class of approaches that tries to circumvent these problems and speed up the computation are so-called warm-starts (Egger et al., 2021; Galda et al., 2021; Truger et al., 2024a). The general idea is to give the algorithm an advantage by utilizing known or efficiently generated results, such as optimized parameter values from related problem instances and classically approximated parameterizations or solutions for the problem at hand. Several techniques can be used to warm-start quantum algorithms (Truger et al., 2024a). In particular, there are two major groups of approaches for warm-starting VQAs. The first group encodes an approximate solution into the algorithm itself. In this sense, Egger et al. propose a QAOA variant called warm-started QAOA (WS-QAOA) for the MAXCUT problem, that uses an approximate solution to prepare a biased initial state and improve upon the approximation (Egger et al., 2021). Similarly, Tate et al. propose a biased initial state directly based on a relaxation of MAXCUT to warm-start the QAOA (Tate et al., 2023). The second group focuses on obtaining

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good initial parameter values that are improved upon in the optimization. One way to get good parameter values is to identify similar problem instances and use their optimized parameters for another instance. This transfer of parameters has been investigated, e.g., by Galda et al. for QAOA (Galda et al., 2021).

This work combines the two approaches and investigates the possibility of parameter transfers between instances of the MAXCUT problem for Egger et al.’s WS-QAOA. Therefore, we analyze the algorithm’s performance on subgraphs of random graphs up to degree 5 to explore regularities and a structure for potential parameter transfers. Moreover, we describe how the insights gained from this analysis are utilized to determine good parameter values for larger graphs when warm-starting is employed. We demonstrate this by finding good initial parameter values for a 50-node random graph instance with a maximum degree of 5 that are very close to the actual optimum. The remaining sections of this paper are organized as follows. Section 2 introduces the necessary background and related work. Section 3 describes how the investigation was performed. The insights and results as well as their utilization are covered in Section 4. Section 5 discusses limitations and possible extension of the method. Finally, Section 6 summarizes our work and gives an outlook on possible future work.

2 BACKGROUND AND RELATED WORK

The QAOA, one of the most prominent examples of VQAs, was introduced in a seminal work by Farhi et al. (2014). They illustrate that the algorithm can be applied to approximate solutions to the MAXCUT problem. In short, the parameterized circuit that is executed on the quantum computer consists of an initialization to a uniform superposition and two subcircuits parameterized by $\hat{\gamma}$ and $\hat{\beta}$, respectively. These subcircuits are referred to as the problem circuit and mixer. The problem circuit is constructed such that its ground state is an optimal solution to the problem at hand, while the mixer’s ground state is the initial state of the overall circuit. The problem circuit and mixer are repeated p times, which is referred to as the QAOA depth of the Ansatz. Thus, the overall circuit implements a trotterized approximation of an adiabatic evolution that aims to achieve a transition from the known ground state of the mixer into the unknown ground state of the problem circuit, representing an optimal solution. The algorithm converges towards an optimum with the increase of the QAOA depth p .

A partitioning of the nodes of an undirected graph into two sets is called a *cut*, with the number of edges between the two sets being the value of the cut. A MAXCUT or Maximum Cut is thus a partitioning that maximizes the number of edges between the two sets. In the case of weighted graphs, the goal is to maximize the sum of the edge weights between both sets. The problem is well known to be NP-hard, but can be approximated. Executing the QAOA for a set of parameters creates a distribution of cuts, with the expected value of the cut being referred to as the energy. The objective of the QAOA is to optimize this energy, thereby producing good cuts.

Egger et al. (2021) adapted the QAOA to utilize approximations to the MAXCUT problem as a starting point by encoding them into the initial state of the quantum circuit. The adaptation, called *Warm-Started QAOA* (WS-QAOA), makes use of the best known classical approximation algorithm for the MAXCUT problem by Goemans and Williamson (1995). Thereby the QAOA is enhanced in the sense that less optimization may be necessary if the initial state is biased towards a good solution as a starting point or, conversely, lower QAOA depth suffices to reach a good result quality, making WS-QAOA a promising algorithm for current NISQ hardware. Specifically, instead of the uniform superposition, each qubit in the initial state of WS-QAOA is more likely to be in either of the states $|0\rangle$ or $|1\rangle$ as prescribed by the approximation obtained from the Goemans-Williamson algorithm (GW). The deviation of the initial state from the uniform superposition and the bias towards the approximation is controlled by a regularization parameter ϵ . In addition, the mixer in WS-QAOA is adapted for the modified initial state. Egger et al. also propose a modified mixer that does not have the initial state as a ground state, however, as a first step we focus on the version that is in line with the adiabatic evolution behind QAOA.

Another way to warm-start VQAs, is by finding a good initialization for the Ansatz parameters, e.g., by pre-computing viable parameter values or transferring optimized parameters from related problem instances (Truger et al., 2024a). The advantage here is, that all VQAs can be warm-started by parameter initializations without having to change the underlying algorithm. The difficulty lies in finding viable parameter values that yield good results. One approach is to optimize parameters for a smaller problem instance and then transferring the optimized parameter values to the actual target problem. To this end, Galda et al. (2021) investigated the transfer of optimal QAOA parameter values for the MAXCUT problem between problem instances. In particular, they

considered regular and random graphs up to a degree of 8 and 6, respectively. This analysis showed that a transfer between instances is possible and primarily depends on the degree of subgraphs. For instance, a transfer between k - and d -regular instances is likely to be successful, provided that k and d are both even or both odd. Our work extends this analysis to WS-QAOA, for which the degree alone is not enough to guarantee a successful transfer anymore.

There are several works similar to the ones introduced above. Tate et al. (2023) successfully warm-start the QAOA for MAXCUT with low-rank Burer-Monteiro relaxations, while Cain et al. (2022) show that a warm-start that naively initializes the circuit with an approximation fails. Shaydulin et al. (2021) introduce a repository for optimized QAOA parameters that enables parameter transfers. Brandao et al. (2018) show that the landscapes of large 3-regular graphs are instance independent, whereas Akshay et al. (2021) demonstrated a concentration of optimal QAOA parameters for $p \in \{1, 2\}$. Shaydulin et al. (2023) explain how to transfer parameters for weighted MAXCUT. More generally, Truger et al. (2024b) document warm-starting patterns for quantum algorithms, including warm-starts via biased initial states and warm-starts via parameter transfers. However, none of the work that we are aware of combines a warm-start via a biased initial state with a parameter transfer.

The *Quantum Modelling Extension* (QuantME) introduces a warm-starting task that conceptually facilitates parameter transfers for WS-QAOA in workflow-based quantum applications (Beisel et al., 2023). In particular, this task provides configuration attributes for warm-starting quantum executions utilizing biased initial states and parameter initializations. However, it does not provide guidance on how to effectively combine the two.

3 PARAMETER TRANSFER FOR WS-QAOA

This section introduces our approach to parameter transfers for WS-QAOA for the MAXCUT problem. We first give an overview of the decomposition of such parameter transfers based on subgraphs, before we provide more details on the concrete experimental setup for our analysis.

3.1 Overview

On a high level, our approach considers the transferability of parameters for subgraphs to derive a param-

eter transfer for a targeted problem instance. As was shown in the original QAOA paper, the energy values for a particular graph can be evaluated as the sum over subgraph contributions (Farhi et al., 2014). We investigate the QAOA at $p = 1$, where only two parameters need to be optimized. To expand the investigation to $p > 1$ one must conduct the analysis on larger and more subgraphs as will be discussed in Section 5. In general, the investigation is centered around energy landscapes of subgraphs. Energy landscapes are created by systematically evaluating the QAOA circuit in its parameter space and computing the corresponding energy values. They allow for the comparison of energy extrema within the parameter space to draw conclusions about the transferability of parameter values.

To determine the transferability of parameters for WS-QAOA, we investigate the energy landscapes for different subgraphs while considering different possible initial states. Specifically, we assess whether optimal parameter values coincide, i.e., whether good values for one instance are also suitable for another. For our purposes, optimal parameters are located at maxima, since we investigate the MAXCUT problem. For a fixed set of parameters, the energy is evaluated according to Equation (1) (cf. Galda et al. (2021)).

$$\begin{aligned} \langle C \rangle_{|\hat{\gamma}, \hat{\beta}\rangle} &= \langle \hat{\gamma}, \hat{\beta} | C | \hat{\gamma}, \hat{\beta} \rangle \\ &= \langle \hat{\gamma}, \hat{\beta} | \frac{1}{2} \sum_{(i,j) \in E} (1 - Z_i Z_j) | \hat{\gamma}, \hat{\beta} \rangle \\ &= \frac{|E|}{2} - \frac{1}{2} \sum_{(i,j) \in E} \langle \hat{\gamma}, \hat{\beta} | Z_i Z_j | \hat{\gamma}, \hat{\beta} \rangle \quad (1) \\ &= \frac{|E|}{2} - \frac{1}{2} \sum_{(i,j) \in E} e_{ij}(\hat{\gamma}, \hat{\beta}) \end{aligned}$$

C in Equation (1) is the problem-specific operator in QAOA and e_{ij} is the contribution of an individual edge in the set of edges E . This means, instead of looking at the graph as a whole, we can break down the problem by investigating how each edge influences the energy. The contribution e_{ij} can be determined by looking at the subgraph that is induced by the edge. The induced subgraph consists of the edge itself, called the *central edge* that connects the *central nodes*, and all nodes (and connecting edges) that are at most distance p away. Therefore, the worst case is that we have to consider as many subgraphs, as there are edges in a graph. In reality, however, many subgraphs appear repeatedly. For example, for k -regular graphs, there are k different subgraphs that need to be investigated. Additionally, in WS-QAOA we need to consider different initializations due to different possible approximations encoded into the initial state as

warm-starts. Thus, the number of energy landscapes that need to be taken into account increases in the context of WS-QAOA compared to standard QAOA.

To facilitate the identification of individual subgraphs we introduce the following notation, which is adapted from Galda et al. (2021): A subgraph is referred to as $(x-y)-z$, where $x \leq y$ are the number of neighbors of the two central nodes and z is the number of neighbors shared by both nodes. For example, the graph in Figure 1 is called $(3-3)-0$.

3.2 Experimental Setup

Assuming a fixed configuration of WS-QAOA, our analysis extends over every possible initialization for every subgraph of regular graphs up to degree 5. After accounting for isomorphisms, this yields 214 different landscapes. Expanding the scope of the investigation to random graphs up to degree 5 yields 35 subgraphs and increases the total number of landscapes to 620. For reasons discussed in Section 4.2, this number can be reduced to 558.

To create landscapes for a subgraph, one first needs to construct the QAOA circuit, where each node corresponds to one qubit. As discussed previously, we adopt WS-QAOA from Egger et al. (2021) in its basic form. The first consideration is the depth p that we choose as $p = 1$ resulting in a circuit with two parameters (γ, β) to be optimized and two-dimensional energy landscapes that can be illustrated. Additionally, WS-QAOA can be adjusted by its regularization parameter $\varepsilon \in [0, 0.5]$, where $\varepsilon = 0.5$ leads to the original QAOA and lower values increase the influence of the approximate solution utilized as a warm-start. To put more emphasis on the warm-start, we choose $\varepsilon = 0.1$. To prepare the biased initial state, a rotation R_Y by an angle θ_i is applied to each qubit q_i , where $\theta_i = 2 \arcsin \sqrt{\varepsilon}$ if the corresponding node

is initialized with 0, and $\theta_i = 2 \arcsin \sqrt{1 - \varepsilon}$ if it is initialized with 1. Thus, a single qubit reproduces its initialization with probability $1 - \varepsilon$ if we were to measure it. For example, a qubit of a node initialized with 0 measures $|0\rangle$ with probability $1 - \varepsilon$. An example for the WS-QAOA circuit is given in Figure 1. After constructing the circuit, it is executed on a simulator where we measure the qubits of the central edge, which are q_0, q_1 in all our experiments. The portion of measurements, where $q_0 \neq q_1$ is the probability of the edge being cut, i.e., the energy contribution e_{ij} of Equation (1). We create the landscape by sampling the energy in the parameters space $(\gamma, \beta) \in [0, 2\pi] \times [0, \pi]$. To this end, we sample 30 parameter values for each dimension to obtain the two sets Γ and B , resulting in a total of $|\Gamma| \cdot |B| = 900$ equidistant samples. Since we are dealing with unweighted graphs, the landscapes are periodic, and it is sufficient to analyze the parameter space within the given range.

4 RESULTS

This section presents the results of the analysis for subgraphs and the transferability of parameters between graphs in WS-QAOA for MAXCUT. In particular, we first look at the landscapes of regular graphs, before moving on to random graphs. Next, we introduce a transferability map to analyze the transferability in more detail. Finally, we utilize the insights to find good parameters for a larger graph instance. For verification and reproducibility, the results can be accessed online (Obst and Truger, 2024). This includes a visualization of all subgraphs with their corresponding landscapes and numeric data.

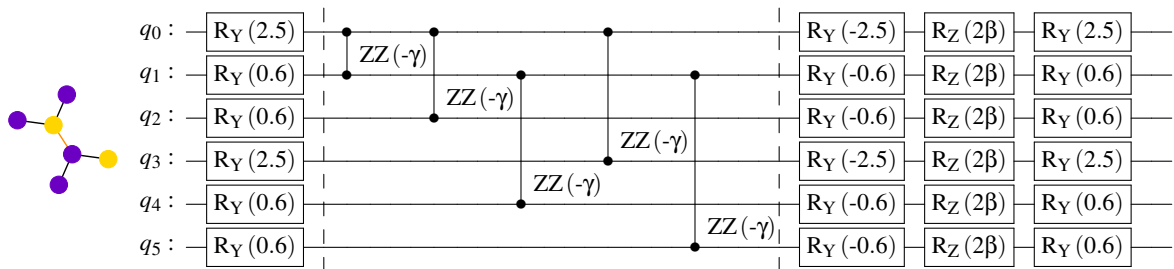


Figure 1: Example subgraph and its circuit consisting of initialization, problem circuit, and mixer separated by dashed lines. The subgraph is depicted with its initialization where purple and yellow nodes correspond to an initialization of 0 and 1 respectively. Each qubit corresponds to one node of the graph, where the first two qubits q_0, q_1 are the central nodes, and each ZZ gate corresponds to an edge in the graph. Since $\varepsilon = 0.1$, the parameters of the initial R_Y gates are $2 \arcsin \sqrt{\varepsilon} \approx 0.64$ if the initialization of the corresponding node is 0, and $2 \arcsin \sqrt{1 - \varepsilon} \approx 2.50$ if the initialization is 1. The parameter values are rounded to one decimal place.

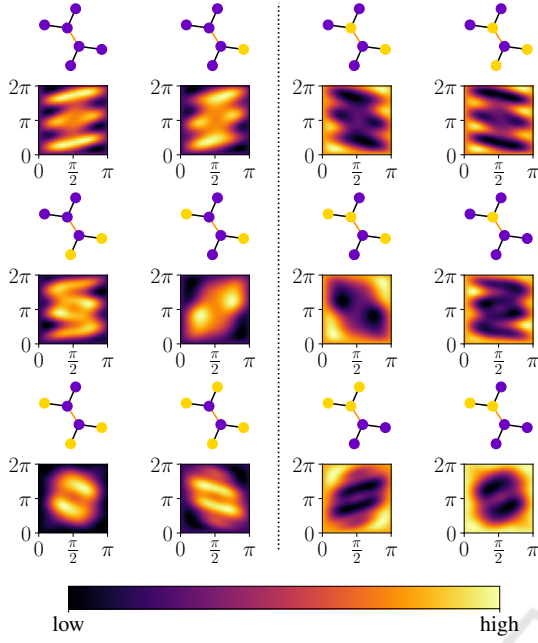


Figure 2: Graph (3-3)-0, one of the three subgraphs of a 3-regular graph, with all its initializations up to isomorphism. An initialization with 0 is represented by a purple node, an initialization with 1 by a yellow node. The central edge that is being measured is highlighted. Values of the landscape are normalized, as we only care about maxima and minima, not their exact value. The graphs are ordered such that all initializations on the left have central nodes that are initialized equally. Additionally, the landscapes are paired with their mirrored version on the opposite side.

4.1 Landscapes for Regular Graphs

In the following, we analyze landscapes for regular graphs and describe regularities found between the landscapes.

4.1.1 3-Regular Graphs

Examining the landscapes for subgraphs on their own helps us gain a better understanding of the reasons for good transferability and identify regularities among the results. First, we focus on the subgraphs of regular graphs, as their regularities also appear in other subgraphs. Figure 2 shows the most general subgraph in the sense that there are no common neighbors among the nodes of the central edge. The graph is depicted with every possible initialization indicated by a color coding: Nodes initialized to 0 are colored in purple, nodes initialized to 1 are colored in yellow. Isomorphisms are not shown, as they produce identical results. Moreover, we also consider it an isomorphism if all initializations of the nodes are inverted, as such configurations would also result in the same energy values. When checking for isomorphisms, it is im-

portant to label the central edge, as it is the one being measured as described in Section 3.2. This makes no difference for d -regular graphs with $d \geq 3$, but is relevant for small random graphs. That way, all 12 different initializations and their corresponding landscapes are presented in Figure 2. All landscapes are normalized for the visualization. The maximum is the brightest spot and the minimum the darkest. For the transfer of parameters, the actual energy values are less relevant than the locations of extrema, particularly the global maxima, as these mark the optimal parameter values. Therefore, we can disregard that scale for transfers and in the visualizations. For a better illustration of regularities among the landscapes, they are grouped in two categories: The first (left) half shows landscapes where outskirts have a low energy, while the center generally shows a high energy. In the second (right) half, it is the other way around. The difference between the two groups lies in the initialization of the central nodes. In the first category, the nodes are initialized identically, i.e., the central edge is not part of the cut. For the second category, the edge is part of the cut, i.e., nodes are initialized differently. Intuitively, this is explained by the fact that we only look at the energy of the central edge. Given a good initialization, parameters that favor the initial state, i.e., parameters around zero, should be expected to result in high energy values.

The chosen arrangement of Figure 2 highlights another relationship between the landscapes. The six landscapes on the right-hand side appear to be inverted mirror images of those on the left-hand side. More precisely, the landscapes on the opposite side have the same shape but appear to be flipped vertically/horizontally and inverted, i.e., minima of one landscape become maxima in the other and vice versa. Landscapes that are related this way also have a connection in the respective graphs: Taking one of the central nodes and inverting the initialization of each adjacent node, including the other central node, results in an isomorphic graph. Future work may provide a better understanding of this phenomenon through a deeper mathematical analysis.

4.1.2 Subgraphs with Common Neighbors

As mentioned before, we can understand the subgraphs $(x-y)-z$ with $z \geq 1$ as a version of the graph $(x-y)-0$ where the central nodes have z common neighbors. Analogous to the observations of Galda et al. (2021), the landscapes of k -regular subgraphs are very similar for a fixed k when their initialization is equivalent, as explained in the following. This fact is demonstrated in Figure 3 with equivalent initializations for the subgraphs (3-3)-0, (3-3)-1,

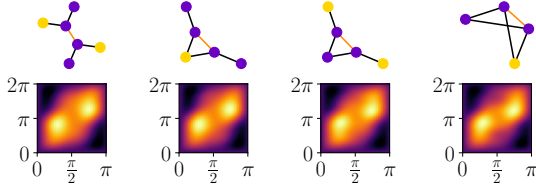


Figure 3: Subgraphs of 3-regular graphs with similar landscapes. From left to right, the subgraphs are (3-3)-0, (3-3)-1, (3-3)-1, (3-3)-2. Similar landscapes arise when identically initialized neighbors are merged into a common neighbor of both central nodes. The MAD between the normalized landscapes is at most 0.03. The second and third landscape look alike, because splitting the common neighbor would result in isomorphic graphs.

and (3-3)-2, which induce resembling landscapes. To quantify this similarity, we determine the *Mean Absolute Difference (MAD)*. For two normalized energy landscapes E_1, E_2 and sets of parameter values Γ, B it is calculated as follows:

$$\text{MAD}(E_1, E_2) = \frac{1}{|\Gamma| \cdot |B|} \sum_{(\gamma, \beta) \in \Gamma \times B} |E_1(\gamma, \beta) - E_2(\gamma, \beta)|$$

Since the energy is in the interval $[0, 1]$, identical landscapes have an MAD of 0 and opposite landscapes have an MAD of 1. In Figure 3, the graph (3-3)-1 is shown twice with different initialization, yet their landscapes look identical. To be precise, their MAD is 0.02. While they are in fact non-isomorphic, the similarity appears because of their relation to (3-3)-0: In both cases, replacing the common neighbor of the central nodes with two identically initialized separate nodes would create the same version of (3-3)-0, i.e., the initialization of these subgraphs is equivalent to the one depicted for (3-3)-0. Hence, (3-3)-0 is the most general of the three subgraphs, as it allows for the largest number of initializations, and “splitting” common neighbors by replacing them with separate nodes returns one of these instances. Since both graphs are descendants of the same parent, we consider only one of them in further analyses.

Increasing z , i.e., introducing common neighbors, does not induce additional initializations to consider over those known for the most general subgraph. Conversely, there are initializations for (3-3)-0 for which there is no equivalent initialization for (3-3)-1 or (3-3)-2. For example, in the top right graph depicted in Figure 2, the neighbors of the central nodes are initialized both with 1 and both with 0, respectively. Thus, no pair of these nodes could be merged into a common neighbor and there is no equivalent initialization for (3-3)-1.

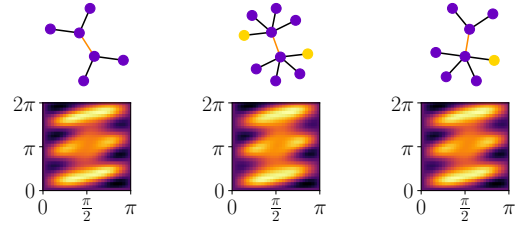


Figure 4: Different types of graphs that produce similar landscapes. Neighbors that are initialized oppositely (one is 0, the other is 1) cancel each other out. Removing them does not substantially affect the landscape, i.e., the MAD between them is smaller than 0.04.

4.1.3 Node Canceling

We briefly extend the analysis to subgraphs of 5-regular graphs, whose landscapes exhibit similarities with those of 3-regular graphs. Namely, all landscapes of graph (3-3)-0 are also present among those for (5-5)-0. An example is depicted in Figure 4. Again, the similarity is due to a connection between the underlying graphs: The central nodes of the (5-5)-0 instance each have two neighbors that are initialized differently. Disregarding such pairs of neighbors and removing them from the graph on both sides of the central edge results in the graph (3-3)-0 which produces the same landscape. In that sense, nodes can cancel each other out and some initializations for (5-5)-0 can be reduced to an equivalent initialization for (3-3)-0. We refer to this as node canceling.

4.2 Random Graphs

The analysis can also be extended to random graphs. In particular, the observations regarding common neighbors and node canceling as introduced above also apply to subgraphs of random graphs up to a degree of 5. There are 35 subgraphs with a total of 620 initializations. However, due to equivalent initializations of graphs with common neighbors, these can be reduced to 558 that need to be analyzed. For subgraphs of random graphs, the number of neighbors of the central nodes can differ. For example, the subgraph (3-5)-0 has a central node that is missing two neighbors compared to (5-5)-0. The rightmost example in Figure 4 illustrates that node canceling also leads to resembling landscapes for these types of subgraphs.

However, subgraphs of random graphs also introduce landscapes with shapes that are not present for regular graphs. Specifically, these landscapes appear when the central nodes have a mixed parity, i.e., one central node has an even and the other has an odd degree. Still, subgraphs of random graphs exhibit the

same regularities as regular graphs, such as the mirrored relationship, merged nodes, and node cancelling. Node cancelling is particularly interesting, as we can also go in the opposite direction. Instead of removing nodes, we can also add them to a graph, until both central nodes have the same number of neighbors. That way, subgraphs of random graphs can be related to those of regular graphs. Clearly, this is only possible if both central nodes have the same parity, i.e., the degree of the central nodes is either even or odd for both, as we add two nodes at a time. Therefore, subgraphs of random graphs fall in one of two categories: Either their central nodes have the same parity, or they have mixed parity.

4.3 Transferability Map

This section examines the parameter transferability between subgraphs of random graphs up to a degree of 5 in more detail. To this end, we create a transferability map that visualizes the quality of such transfers. We describe the construction of the transferability map, the key observations, and the influence of the degree of central nodes on the transferability.

4.3.1 Constructing the Transferability Map

To quantify the transferability between different subgraphs we determine a *transferability coefficient* τ . It is calculated by taking the sampled optimal parameters of one subgraph, the donor graph, and evaluating their performance for another subgraph, the acceptor graph. Determining the transferability is straightforward given the sampled landscapes. The transferability coefficient τ can be computed as follows:

$$\tau = \text{avg}_{\gamma_{\max}, \beta_{\max}} \left(\frac{E_{\text{acc}}(\gamma_{\max}, \beta_{\max})}{\max_{\gamma, \beta} \{E_{\text{acc}}(\gamma, \beta)\}} \right) \quad (2)$$

We take the locations of global maxima in the sampled landscape, $(\gamma_{\max}, \beta_{\max})$, of the donor subgraph (there can be multiple maxima), evaluate the energy E_{acc} of the acceptor graph at these locations, normalize by the maximum energy of the acceptor graph and average the values. The coefficient τ is a value between 0 and 1, where 0 is the worst possible transferability and 1 is the optimal transferability.

Figure 5 shows the transferability coefficient for every pair of two subgraphs and every initialization. As Galda et al. noted, transferability is a directional property. Therefore, the map is not symmetric along its diagonal axis. The appearance of the map depends on the arrangement of the subgraphs and their initializations. While there are multiple possibilities to order the graphs and their initializations, we decided for

an arrangement that is beneficial for the illustration of some key observations. In particular, the graphs are ordered by increasing minimum degree x and increasing maximum degree y , i.e., first all the graphs $(1-*)$, then $(2-*)$, etc. In addition, within each category, the number of merged nodes z increases. The initializations of a single subgraph are divided into two groups, the first having identically initialized central nodes, the second having central nodes initialized oppositely. The complete ordering of initializations of a subgraph can be found online (Obst and Truger, 2024) together with all landscapes.

4.3.2 Key Observations

As the transferability map contains a substantial amount of information, we draw attention to key observations that unveil its utility. For one, the map facilitates finding a suitable donor graph for a given subgraph by scanning the respective row to identify good candidates for a parameter transfer. Moreover, the map helps to identify certain patterns and regularities, to estimate the success of parameter transfers, and to gain insights into properties that can be used to make more general statements about transferability.

An important observation is, that the initialization introduced with WS-QAOA significantly influences the transferability. To be precise, the transferabilities described by Galda et al. (2021) no longer apply when a biased initial state is introduced, as these conditions relied on graph structure alone. For example, optimal parameters of a subgraph were trivially transferable to the same subgraph. However, for the same instance in WS-QAOA with different initializations, such transfers may fail. In particular, this is the case across initializations where the central edge is cut and others where the central edge is not cut, i.e., the two groups described earlier. This restriction is visible in the map by the fact that almost all subgraphs have four distinct sectors that appear like a checkerboard pattern (see Figure 5 box A). Note that such a section contains all possible initializations for a single subgraph. The four sectors reflect all possible transfers for the different initializations of the central nodes of the donor and acceptor graph. They illustrate, that a transfer is most successful, when the central nodes are either initialized identically in both graphs or initialized differently in both.

The next observation is, that the number of common neighbors has little influence on the transferability (B), which is reflected in the similarity of sections separated by dotted lines in Figure 5. This occurs because splitting common neighbors results in very similar landscapes, as discussed above and shown in Figure 3. This phenomenon is already present when no

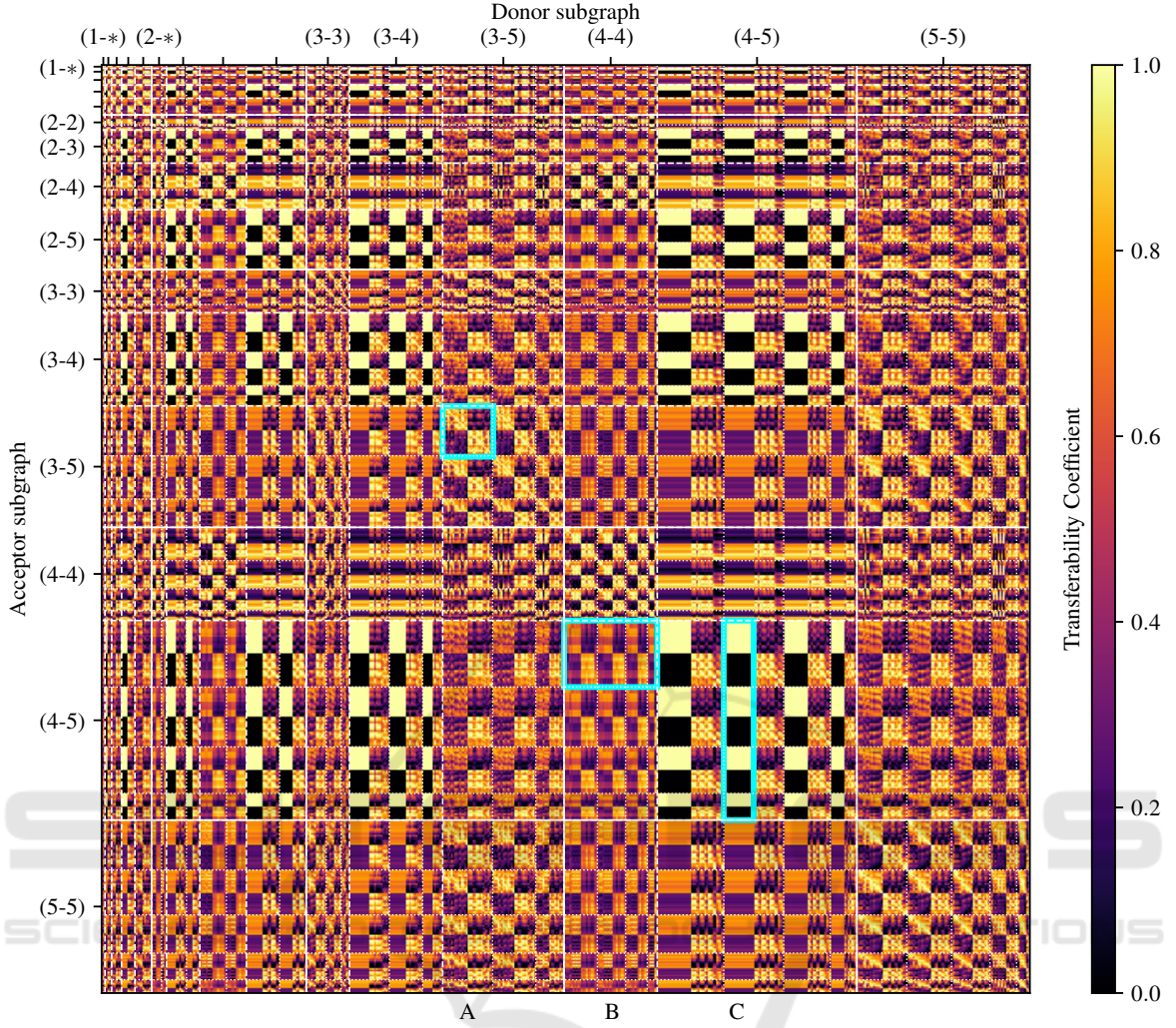


Figure 5: *Transferability coefficients* between every pair of subgraphs forming the *Transferability Map*. Each pixel is a coefficient $\tau \in [0, 1]$. Dark colors indicate a bad transferability, bright colors indicate a good transferability. Each row is an acceptor graph, each column a donor graph. Thick white lines distinguish graphs with differing minimum degree. Dashed white lines separate graphs with different degrees of the central nodes, dotted lines further divide the number of common neighbors. Overall, there are 35 subgraphs with a total of 558 initializations, starting with low degrees up to degree 5. For clarity, some of the labels are omitted: (1-*) stands for (1-1) through (1-5), (2-*) stands for (2-2) through (2-5). The highlighted sections and labels A, B, C serve as references for the observations described in Section 4.3.2.

biased initial state is involved (Galda et al., 2021). For our purposes, this means that we need to prioritize the initialization over the exact type of graph when looking for a suitable donor graph.

Distinctive features of the map are absolutely perfect and disastrous transferabilities in some cases (C). These become apparent as yellow and black patches in Figure 5, most notably in the sections of (4-5). Here we have optimal and worst case transferability with coefficients of 1 and 0, respectively. This is due to the shape of the respective landscapes, which exhibit either exactly one maximum or one minimum that is located right at the center of the landscape, re-

sulting in either a perfect transfer or the worst possible. Interestingly, these patches only appear for donors whose central nodes are initialized the same. However, the other sector, where the central nodes of donor and acceptor are initialized oppositely, also exhibits relatively high coefficients. This sector is particularly important for a parameter transfer of WS-QAOA, as we will discuss in Section 4.4.

4.3.3 Influence of Node Degrees

Another important characteristic is the degree of the central nodes, particularly, their parity. Galda et al. already described this and noted that a transfer between

k -regular and d -regular graphs is likely to be successful, provided k and d are either both even or both odd (Galda et al., 2021). When taking the initialization of the central edge into account, this also seems to hold with warm-starts in place. For example, (5-5) appears to be a better donor for (3-3) than (4-4).

They also found that there is a good transferability between graphs $(j-k)$ and $(l-m)$ provided that $\{j, k, l, m\}$ are all odd or all even. This also seems true for WS-QAOA. For example, (3-5) as an acceptor has brighter sections when receiving parameters from (5-5) rather than (4-4). Conversely, a transfer from odd to even parity is also unlikely to be successful. Notably, for (4-4) as an acceptor, the graph (2-4) appears to be a better donor than (3-3). Interestingly, graphs with even degrees do not show a checkerboard pattern when receiving parameters from mixed parity donors. An example for this would be (4-4) accepting parameters from (4-5).

Moreover, Galda et al. showed cases of good transferability between subgraphs with mixed parity. With WS-QAOA, there seems to be good transferability if both donor and acceptor graph have a mixed parity, since the aforementioned sections with yellow and black patches appear in exactly these cases.

4.4 Application for Transferability

This section describes, how we can use the above insights to obtain viable parameter values for larger graphs composed of subgraphs. To this end, we decompose a larger graph into its subgraphs and try to reduce the number of subgraphs that we have to consider. We illustrate this application by determining viable parameter values for the 50-node random graph with 75 edges and a maximum degree of 5 shown in Figure 6. A suitable initialization for the MAXCUT problem on this graph was determined using the GW algorithm (Goemans and Williamson, 1995).

The following procedure is based on the fact, that the total energy is composed of contributions of the

subgraphs as described in Section 3.1. We can leverage the insights described above to obtain an approximation of the landscape of the target graph shown in Figure 6, from which we can derive a good parameterization. Specifically, subgraphs can be transformed into a standardized format to increase the number of isomorphic subgraphs, thereby reducing the number of subgraphs that need to be considered and minimizing the effort required to determine good parameter values. The first simplification is to consider only subgraphs of the form $(x-y)-0$ by splitting common neighbors of the central nodes. Since the landscapes are very similar before and after the split, we do not lose much information. Further, a graph falls in one of three categories: Both central nodes have even degrees, both have odd degrees or their degrees are mixed. To transform these subgraphs further, we can utilize the node canceling effects, i.e., add and remove nodes that cancel out. In case that both central nodes have even degrees, we can add two oppositely initialized nodes at a time to a central node until the graph takes the form of (4-4). If their degrees are odd, we can add as many pairs as necessary to end up with (5-5), and for mixed parity we can transform them into instances of (4-5). In total, these 3 graphs allow for 90 different initializations.

Adding up the energy values of the subgraphs with the respective initializations for each sampled point results in the true energy landscape of the target graph. This true energy landscape is illustrated on the left-hand side of Figure 6. Transforming the subgraphs before applying this procedure generates the approximate landscape next to it. The target graph consists of 38 different subgraphs, with the 3 most frequent appearing 8, 7, and 5 times, respectively. The simplifying transformations halve the number of different subgraphs to 19, with the 3 most frequent ones appearing 11, 10, and 9 times, respectively. These 3 most frequent subgraphs produce the landscape shown on the right-hand side of Figure 6.

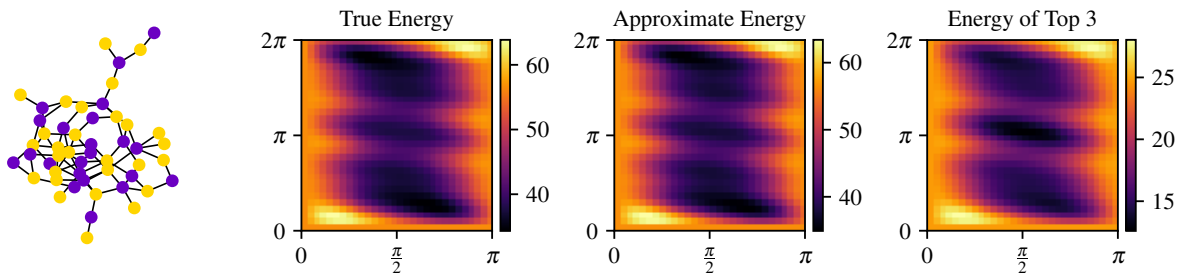


Figure 6: Approximating the landscape of the graph on the left. The 50-node graph is pictured with its initialization, calculated by the GW algorithm. Landscapes from left to right are the true energy landscape, its approximation after transforming all the subgraphs to an equivalent of the form (4-4), (4-5), or (5-5), and the approximation of the landscape by only considering the 3 most frequent graphs after the simplifying transformations.

As a first observation, the landscapes show low energy values in the center and high energy values near the edges. Thus, they fall into the second group of landscapes, where the central edge is cut (cf. Section 4.1.1). This is expected because the initialization is derived from a good approximation utilized for the warm-start, resulting in subgraphs where many central edges are already cut, i.e., initialized oppositely. Further, it is expected that most of the subgraphs will have many such edges, similar to the top right graph of Figure 2. Therefore, the overall landscape resembles the landscape of these subgraphs. Thus, an even simpler transfer would be taking optimized parameters from one of these graphs.

Evidently, the landscapes of Figure 6 appear very similar despite the simplifications described above. In particular, the MAD of true and approximate energy is smaller than 0.02. If we consider only the three most frequent graphs, the landscape changes noticeably, but not substantially. Since we only considered the energy of 30 subgraphs for the latter, the total energy values are generally lower than in the original landscape. However, it is not the total energy value that is crucial for obtaining good parameter values as mentioned in Section 4.1.1, but rather the location of the maxima, that need to remain close. Keep in mind, that the parameters (γ, β) were sampled in the range of $[0, 2\pi] \times [0, \pi]$ with 30 samples per dimension. For the first approximation, the maximizing parameters are off by only one sample in the direction of β , i.e., we are off by $\frac{\pi}{30}$ from the sampled maximum. Surprisingly, the second approximation from the top three subgraphs retains the maximum at the exact same sample position as the true energy.

What the experiments have shown is, that we may only focus our attention towards subgraphs that have a central edge that is cut, as these are the most frequent ones. Additionally, it can suffice to consider only three types of graphs, one for each of the categories of even, odd, and mixed parity.

5 DISCUSSION AND LIMITATIONS

While we analyzed many different subgraphs, and demonstrated how to successfully transfer parameters, there still remain open questions. Here, we discuss limitations of our study and how these shortcomings may be addressed.

For one, it is not always feasible to calculate the entire landscape, e.g., when larger QAOA depths $p > 1$ are involved. In such cases, it is beneficial to resort to the simplifications introduced above, identify

the most frequent subgraphs and optimize the energy of their central edges. The parameter values can then be used as an initialization for the instance at hand. We suspect, that there is a concentration of good parameters in proximity to the optimal parameters of our example. However, further research is required to verify whether this is the case.

As mentioned in Section 3.2, the choice of $\epsilon = 0.1$ puts more emphasis on the initialization. However, this parameter can also be selected differently or even optimized, depending on the objective function in use (Truger et al., 2022). Such alternative objective functions for WS-QAOA may lead to different results for the landscapes and regularities for parameter transferability.

The number of possible initializations grows rapidly with the size of the graph. For this reason, we only analyzed graphs up to degree 5, which may be sufficient for some use cases. While we suspect that the regularities described above generalize to larger degrees, it is unclear whether they actually hold. It would be interesting to see the transferability in WS-QAOA for graphs up to degree 8 as analyzed by Galda et al. Since many of their findings for standard QAOA are consistent with ours for WS-QAOA, we suspect that the regularities also hold for degrees up to 8. However, an analysis could provide further insights into the impact of the degree on transferability.

To calculate the transferability coefficient, we sampled landscapes, which are limited to the grid of points that were evaluated. Therefore, there may be even better parameterizations than those corresponding to the maxima in the sampled landscapes, which could be obtained by additional optimization. Nevertheless, the sampled landscapes appear smooth and thus sufficient for the analysis of parameter transfers.

As noted above, our analysis is centered on a QAOA depth of $p = 1$. While we suspect that some of the regularities may also be present for $p > 1$ and that techniques for determining good parameters could still be applied, further study is required for confirmation. Such an investigation would involve larger subgraphs, since edges that are distance p away from the central edge form the subgraph and contribute to its energy. This implies that more types of subgraphs need to be analyzed, and more initializations have to be considered. The rapid growth of initializations that comes with increasing subgraph size may render an analysis as thorough as for $p = 1$ impractical. More than two circuit parameters further complicate an investigation, requiring the sampling of significantly more data points and limiting the visualization.

Thus far, our analysis is limited to unweighted graphs. Investigating weighted graphs requires a dif-

ferent approach, as the enumeration of every subgraph is impossible with weights. Previous work explored a transfer of QAOA parameters for the MAXCUT problem on weighted graphs (Shaydulin et al., 2023). It is unclear how weighted graphs affect the transferability of optimal parameters for WS-QAOA.

Our analysis was conducted based on noiseless quantum simulation, however, using error-prone NISQ hardware may pose additional challenges and potentials for the approach of parameter transfers in (WS-)QAOA. In particular, it was shown that applying the *circuit cutting* concept to QAOA for MAXCUT by breaking down the graph at hand can significantly reduce the effects of noise on NISQ devices (Bechtold et al., 2023). Circuit cutting focuses on partitioning circuits into multiple smaller circuits, whose individual execution results are post-processed to obtain the original circuit’s result. On a higher level, the approach of parameter transfers for WS-QAOA based on subgraph decomposition is similar to circuit cutting, since the subgraph decomposition in our approach also results in an effective breakdown on the circuit level. Therefore, parameter transfers for (WS-)QAOA based on subgraphs likely allow for similar advantages on current hardware.

6 SUMMARY AND FUTURE WORK

In this paper, we investigated a combination of two warm-starting techniques for the QAOA applied to the MAXCUT problem. The techniques in question are starting the computation with a biased initial state following Egger et al.’s WS-QAOA (Egger et al., 2021) and initializing parameters by transfers from subgraph instances (Galda et al., 2021). To this end, we analyzed energy landscapes of subgraphs of random graphs up to a degree of 5. We documented a number of regularities among the graphs and their initializations that can be leveraged to derive suitable initial parameters for larger graphs. The application of these insights was demonstrated on a 50-node graph instance initialized with a suitable approximation encoded into the biased initial state. We found, that a few small subgraph instances suffice to acquire good initial parameter values for the target graph.

For future work, it would be interesting to increase the scope of the approach, e.g., to graphs of higher degrees or increased QAOA depth p to see whether the regularities continue and to observe potential new ones. Further, some questions remain open, for example why the landscapes of two categories exhibit a mirror relationship. Additionally, tackling different

algorithms or variations like the modified mixer proposed by Egger et al. (2021) would be an interesting expansion. Another idea would be the investigation of similar parameter transfers for WS-QAOA in the context of the MAXCUT problem for weighted graphs, as discussed in Section 5.

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